Name: Mayuri Hirade
Roll No.:13163

In [11]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [20]:
```
url   "https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data"
column_names = ["sepal_length", "sepal_width", "petal_length", "petal_width", "species ]
df   pd.read_csv(url, header=None, names=column_names)
```

In [22]:
```
df
```

Out[22]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [24]:
```
print(df.head())
```

```
   sepal_length  sepal_width  petal_length  petal_width      species
0           5.1          3.5           1.4          0.2  Iris-setosa
1           4.9          3.0           1.4          0.2  Iris-setosa
2           4.7          3.2           1.3          0.2  Iris-setosa
3           4.6          3.1           1.5          0.2  Iris-setosa
4           5.0          3.6           1.4          0.2  Iris-setosa
```

In [32]:
```
df.describe()
```

Out[32]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [34]:
```
df.describe(include   'object )
```

Out[34]:

	species
count	150
unique	3
top	Iris-setosa
freq	50

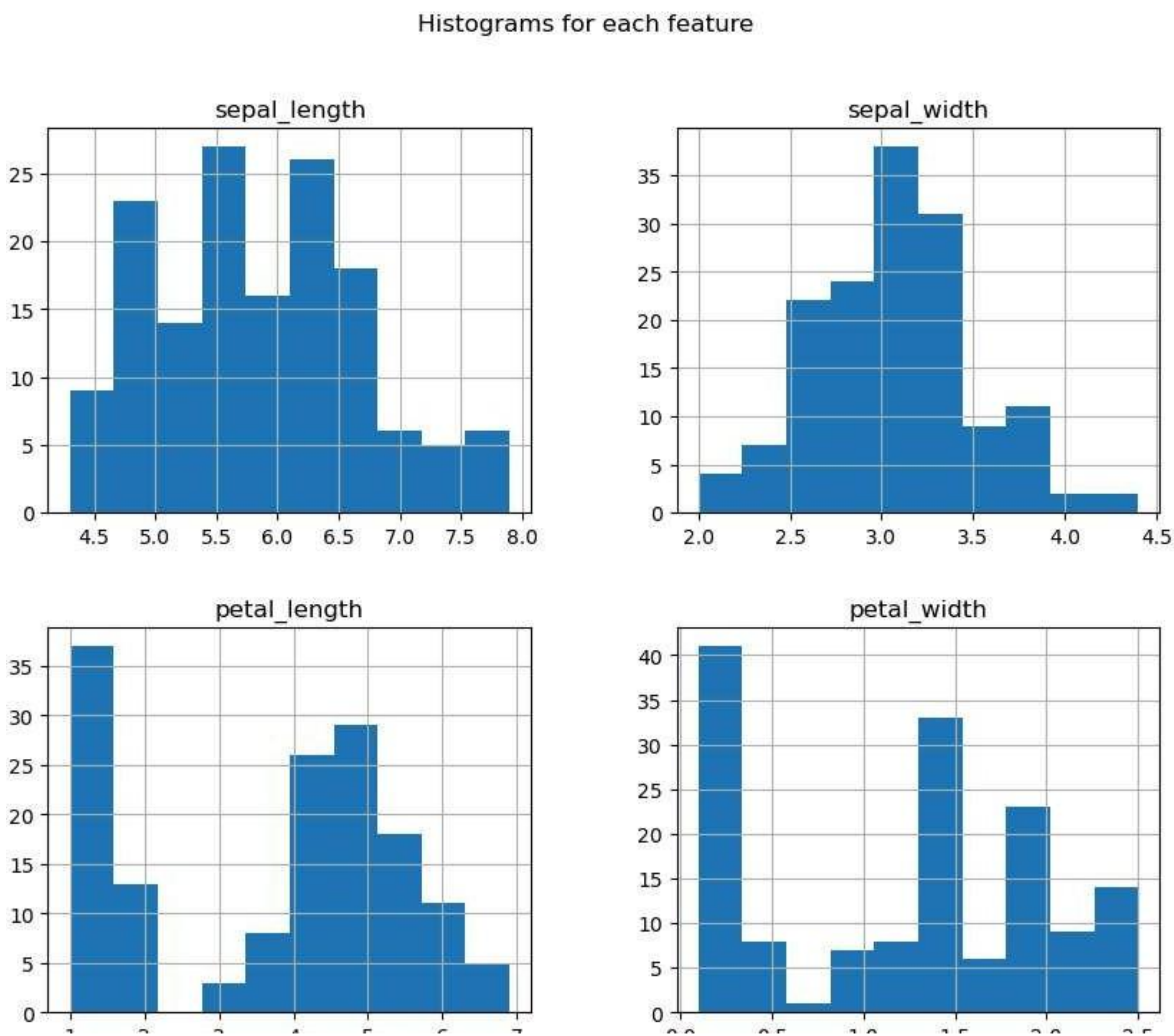
In [36]:
```
df.isnull().sum()
```

```
Out[36]: sepal_length    0
         sepal_width     0
         petal_length    0
         petal_width     0
         species         0
         dtype: int64
```

```
In [40]: print("\n\nThe feat     in the dataset        follows     )
         print("1. Sepal length : ", df 'sepal_length'].dtype)
         print("2. Sepal width : ", df['sepal_width'].dtype)
         print("3. Petal length   ", df['petal_length'].dtype)
         print("4. Petal width : ", df['petal_width'].dtype)
         print("5. Species    , df['species'].dtype)
```

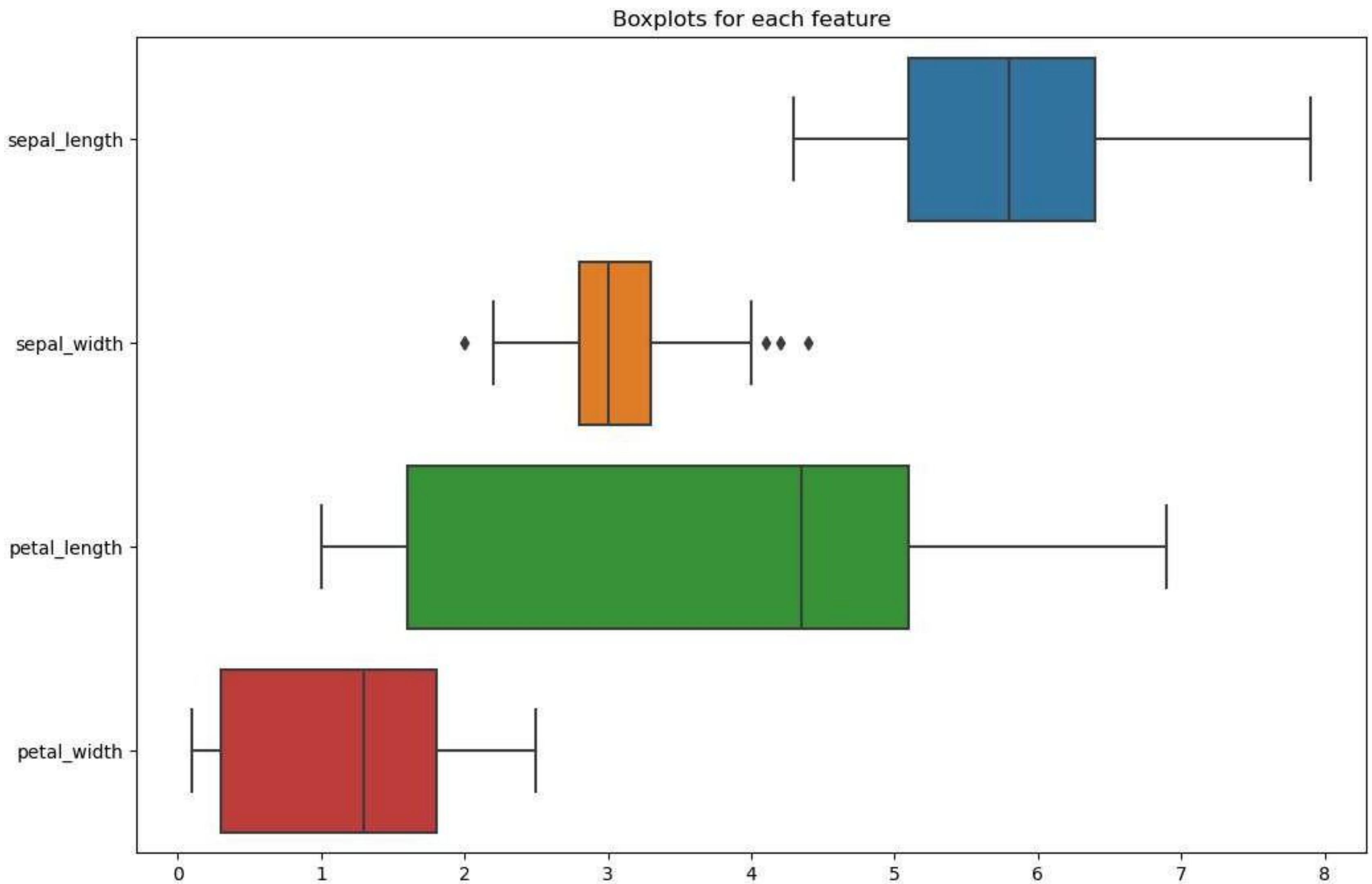
```
The features in the dataset are as follows :
1. Sepal length :  float64
2. Sepal width :  float64
3. Petal length :  float64
4. Petal width :  float64
5. Species :  object
```

```
In [26]: df.drop('species', axi =1).hist(bins=10, figsize=(10  8))
         plt.suptitle("Histograms for each feature")
         plt.show()
```


Histograms for each feature
sepal_length
sepal_width
petal_length
petal_width

1 2 3 4 5 6 7 0.0 0.5 1.0 1.5 2.0 2.5

In [28]:
```
plt.figure(figsize=(12, 8))
sns.boxplot(data=df.drop('species', axi =1   orient=    )
plt.title('Boxplots for each feature')
plt.show()
```



```
In [46]: # Calculate IQR for each feature
         Q1   df.drop('species', axis=1).quantile(0.25)
         Q3   df.drop('species', axis=1).quantile(0.75)
         IQR = Q3 - Q1
         # Define outliers as values outside       IQR
         outliers = ((df.drop('species', axis=1  < (Q1 - 1.5 * IQR)) | (df.drop('species', axis=1  >  Q3   1.5 * IQR
         print(outliers.sum())  # This wi   gi   the number of outliers in each column
```

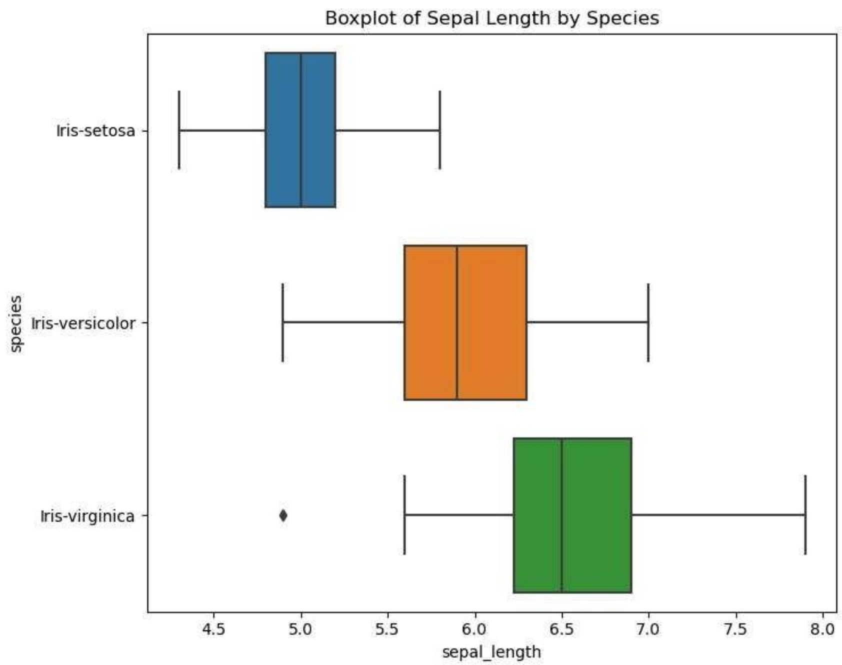
```
sepal_length    0
sepal_width     4
petal_length    0
petal_width     0
dtype: int64
```

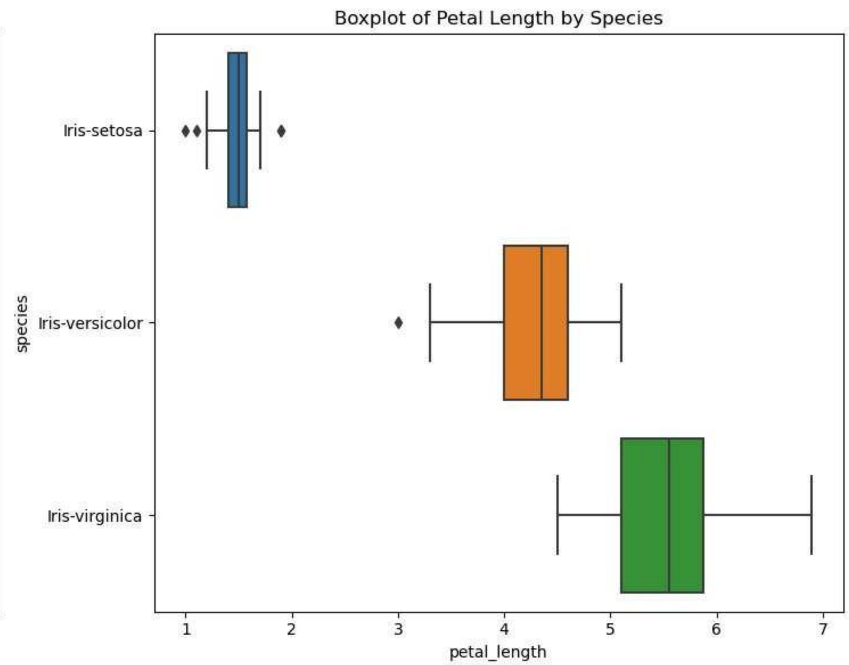
```
In [68]: # Create   figure with       and           for the boxplots
         fig  axes   plt.subplots(1,  , figsize= 15  6))  # 1 row, 2 columns

         # Create the first boxplot for 'sepal_length'    'species'
         sns.boxplot(x='sepal_length', y='species', data=df, ax=axes 0])
         axes[0].set_title('Boxplot of Sepal Length by Species')

         # Create the second boxplot for 'petal_length'    'species'
         sns.boxplot(x='petal_length', y='species', data=df, ax=axes 1])
         axes[1].set_title('Boxplot of Petal Length by Species')

         # Adjust layout for better spacing between the  lots
         plt.tight_layout()
         plt.show()
```


Boxplot of Sepal Length by Species
Iris-setosa
Iris-versicolor
Iris-virginica
species
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
sepal_length


Boxplot of Petal Length by Species
Iris-setosa
Iris-versicolor
Iris-virginica
species
1
2
3
4
5
6
7
petal_length